

Common Errors in Statistics: p-Values

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- Amgen was only able to replicate 11% of the academic literature they studied (Wikipedia, 2024)
- Bayer only replicated 20% of the academic literature they tried (Wikipedia, 2024)
- **Mitch's conjecture:** The replication crisis can be explained, at least in part, by the systemic misuse of statistics in the medical research literature, in particular *p-values*.
- Psychology, economics, and water management are also experiencing a replication crisis for similar yet different reasons (Wikipedia, 2024)

<https://onlinelibrary-wiley-com.proxy.bib.uottawa.ca/doi/book/10.1002/9781118360125>

P-VALUES

The p-value is not the probability that the null hypothesis is true.—Yoccoz [1991]

Before interpreting and commenting on p -values, it is well to remember that in contrast to the significance level, the p -value is a random variable that varies from sample to sample. There may be highly significant differences between two populations and yet the samples taken from those populations and the resulting p -value may not reveal that difference.

Consequently, it is not appropriate for us to compare the p -values from two distinct experiments, or from tests on two variables measured in the same experiment, and declare that one is more significant than the other.

If we agree in advance of examining the data that we will reject the hypothesis if the p -value is less than 5%, then our significance level is 5%. Whether our p -value proves to be 4.9% or 1% or 0.001%, we will come to the same conclusion. One set of results is not more significant than another; it is only that the difference we uncovered was measurably more extreme in one set of samples than in another.

Note that, after examining the data, it is unethical to alter the significance level or to reinterpret a two-tailed test as if one had intended it to be one-tailed.

p -values need not reflect the strength of a relationship. Duggan and Dean [1968] reviewed 45 articles that had appeared in sociology journals between 1955 and 1965 in which the chi-square statistic and distribution had been employed in the analysis of 3×3 contingency tables and compared the resulting p -values with association as measured by Goodman and Kruskal's gamma. Table 8.4 summarizes their findings.

p -values derived from tables are often crude approximations, particularly for small samples and tests based on a specific distribution. They and the stated significance level of our test may well be in error.

The vast majority of p -values produced by parametric tests based on the normal distribution are approximations. If the data are "almost" normal, the associated p -values will be almost correct. As noted in Chapter 6, the

stated significance values for Student's t are very close to exact. Of course, a stated p -value of 4.9% might really prove to be 5.1% in practice. The significance values associated with the F -statistic can be completely inaccurate for nonnormal data (1% rather than 10%). And the p -values derived from the chi-square distribution for use with contingency tables also can be off by an order of magnitude.

The good news is that there exists a class of tests, the permutation tests described in Chapter 6, for which the significance levels are exact if the observations are independent and identically distributed under the null hypothesis or their labels are otherwise exchangeable.

Regardless of which test one uses, it is the height of foolishness to report p -values with excessive precision. 0.06 and 0.052 are both acceptable, but 0.05312 suggests you have let your software do your thinking for you.

This paper started life as an attempt to defend p -values, primarily by pointing out to theoreticians that there are more things in the clinical trials industry than are dreamed of in their lecture courses and examination papers. I have, however, been led inexorably to the opposite conclusion, that the current use of p -values as the "main means" of assessing and reporting the results of clinical trials is indefensible.—P. R. Freeman [1993, 6, p. 1443]

The overall conclusion is that P values can be highly misleading measures of the evidence provided by the data against the null hypothesis.—J. O. Berger and T. Sellke [1987, 7, p. 112]

CONFIDENCE INTERVALS

If p -values are misleading, what are we to use in their place? Jones [1955, p. 407] was among the first to suggest that

an investigator would be misled less frequently and would be more likely to obtain the information he seeks were he to formulate his experimental problems in terms of the estimation of population parameters, with the establishment of confidence intervals about the estimated values, rather than in terms of a null hypothesis against all possible alternatives.

See, also, Gardner and Altman [1996] and Poole [2001].

Confidence intervals can be derived from the rejection regions of our hypothesis tests, whether the latter are based on parametric or nonparametric methods. Suppose $A(\theta)$ is a $1 - \alpha$ level acceptance region

TABLE 8.4. p -value and association

p -value	Gamma		
	<.30	.30 to .70	>.70
<0.1	8	11	5
.05	7	0	0
>.10	8	0	0

- “*p-value is a random variable that varies from sample to sample. [...] it is not appropriate for us to compare the p-values from two distinct experiments, or from tests on two variables measured in the same experiment, and declare that one is more significant than the other.*” (Good, Hardin, 2012, page 155)
- there is no “ordering relationship” between p-values, ie. a p-value is NOT an *order statistic*
- every replication of an experiment will produce a different random ordering
- sorting experimental results by p-value is actually a *random shuffle* of the data

- sorting by p-value and then taking the top 10 gives a *random subset* that will be different for each experiment, or each replica within an experiment
- Google Scholar search for “sorted by p-value” gave 2050 results. Any conclusions drawn in these publications from the ordering of p-values, or which p-values are “most significant”, are **statistically invalid** and by definition (of the p-value as a random variable) **not reproducible**
- find another meaningful metric to sort by within the context of your experiment, otherwise just report alphabetically

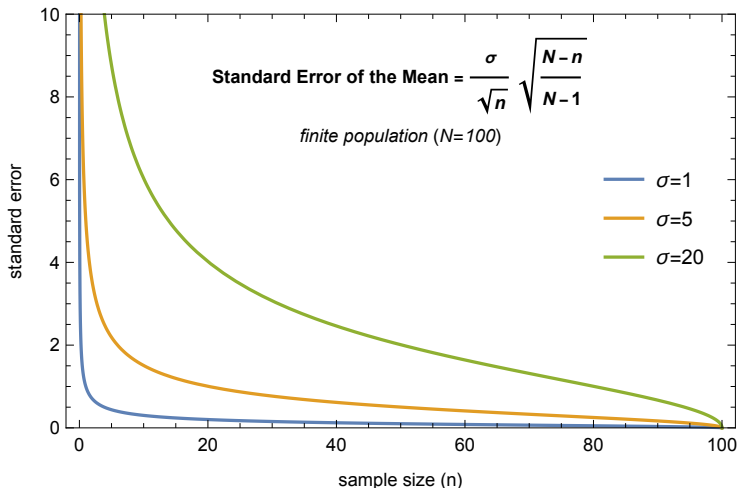
Table 1 Representative marker genes of each fibroblast subpopulation.

	Gene	Fold-change	% cells in cluster	% cells not in cluster	Adjusted p-value
Secretory-reticular	WISP2	15.06	0.83	0.074	0
	SLPI	11.64	0.729	0.065	0
	CTHRC1	7.69	0.759	0.092	0
	MFAP5	0.84	0.474	0.057	1.29E-259
	TSPAN8	4.28	0.569	0.056	3.44E-107
Pro-inflammatory	CCL19	12.51	0.343	0.096	3.79E-75
	APOE	8.484.70	0.868	0.281	3.59E-275
	CXCL2	4.61	0.698	0.39	2.13E-80
	CXCL3	4.35	0.525	0.238	3.77E-63
	EFEMP1	3.12	0.564	0.126	6.36E-167
Secretory-papillary	APCDD1	6.03	0.78	0.11	0
	ID1	3.81	0.60449	0.187	1.80E-109
	WIF1	3.74	0.438	0.035	3.01E-232
	COL18A1	2.96	0.581	0.168	1.68E-113
	PTGDS	2.94	0.559	0.196	2.05E-152
Mesenchymal	ASPN	8.75	0.666	0.067	7.31E-291
	POSTN	5.44	0.620	0.104	2.46E-170
	GPC3	3.58	0.513	0.063	2.83E-177
	TNN	3.42	0.337	0.007	2.10E-286
	SFRP1	3.26	0.406	0.040	5.61E-165

The table shows the five genes selected as marker genes for each fibroblast subpopulation according to their fold-change and enriched expression in comparison with the other subpopulations.

- Reporting p-values like 1.29×10^{-259} is “the height of foolishness” (Good, Hardin, 2012, page 156)
- By comparison, there are $\sim 10^{80}$ protons, neutrons, electrons, neutrinos in the universe
- Sample size required to estimate accurately that far into the tail does not exist (at least not in this universe)
- no mention of p-values in the *Peer Review File* for this article

“I am glad to hear from you. For your question below, it is not quite in my area. But I have some general remarks about something like 10^{-50} . This is the first time for me to see such a small value in a statistical context. In probability we can talk about rare events, and in physics there can be really weak effect. But they are never close to something like 10^{-50} ! Cheers, Minyi” (Minyi Huang, Professor of Statistics, Carleton University, email communication, 2023)

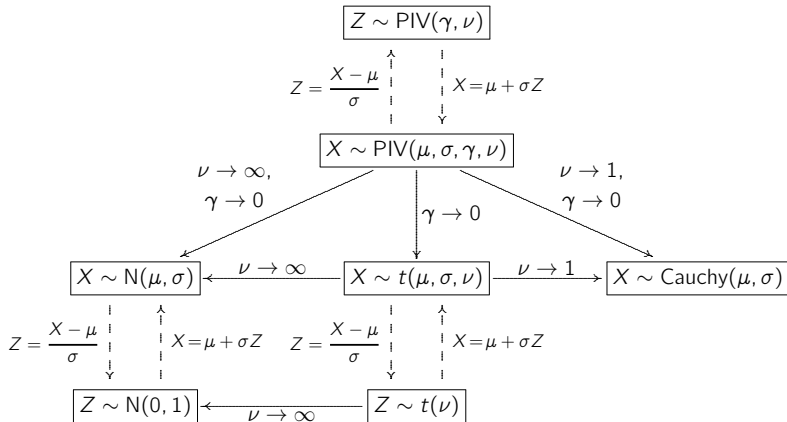


- sampling error is infinite with a sample size of 0
- as sample size n approaches population size N , sampling error converges to 0
- the larger the variability of a population, the bigger sample size required

$$\mu + \sigma N(0, 1) \sim N(\mu, \sigma), \quad \frac{N(\mu, \sigma) - \mu}{\sigma} \sim N(0, 1)$$

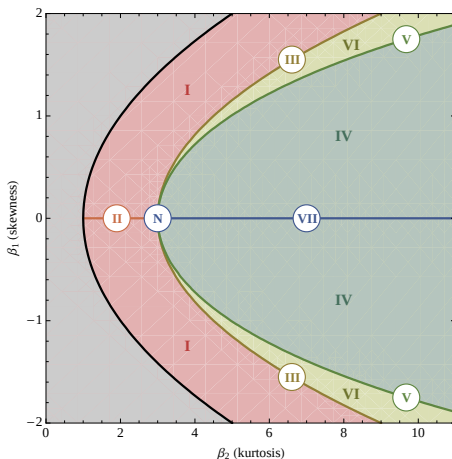
$$N(\mu_1, \sigma_1) + N(\mu_2, \sigma_2) \sim N\left(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2}\right)$$

$$e^{N(\mu, \sigma)} \sim \text{LogNormal}(\mu, \sigma), \quad \text{LogNormal}(\mu_1, \sigma_1) - \text{LogNormal}(\mu_2, \sigma_2) \sim \text{PIV}(\mu_3, \sigma_3, \gamma_3, \nu_3)$$



Pearson (1895) distributions are any valid solution of the differential equation

$$\frac{d \ln f(x)}{dx} = \frac{f'(x)}{f(x)} = -\frac{x + a}{b_0 + b_1 x + b_2 x^2}$$



Type	Support	Distribution(s)
Normal Distribution		
<i>limit of types I, II, III, IV, V, VI and VII</i>		
Main Types		
I	(a, b)	Beta
IV	$(-\infty, \infty)$	Pearson IV
VI	(a, ∞)	Lognormal, F, Beta Prime
Transitional Types		
II	$(-a, a)$	<i>symmetric type I</i> , eg. Uniform
III	$(-a, \infty)$	χ^2 , Gamma, Exponential
V	$(0, \infty)$	Inverse χ^2 , Inverse Gamma
VII	$(-\infty, \infty)$	<i>symmetric type IV</i> , eg. Student <i>t</i> , Cauchy

univariate pearson iv

$$f(x \mid \mu, \sigma, \gamma, \nu) = \frac{C(\gamma, \nu)}{\sigma} \left(1 + \left(\frac{x - \mu}{\sigma \sqrt{\nu}} \right)^2 \right)^{-\frac{\nu+1}{2}} e^{\gamma \tan^{-1} \left(\frac{x - \mu}{\sigma \sqrt{\nu}} \right)}$$

bivariate pearson iv

$$f(x, y \mid \mu_1, \mu_2, \sigma_1, \sigma_2, \rho, \gamma_1, \gamma_2, \nu) = \frac{C_2(\rho, \gamma_1, \gamma_2, \nu)}{\sigma_1 \sigma_2} \frac{e^{\gamma_1 \tan^{-1} \left(\frac{x - \mu_1}{\sigma_1 \sqrt{\nu}} \right) + \gamma_2 \tan^{-1} \left(\frac{y - \mu_2}{\sigma_2 \sqrt{\nu}} \right)}}{\left(1 + \frac{\left(\frac{x - \mu_1}{\sigma_1 \sqrt{\nu}} \right)^2 - 2\rho \left(\frac{x - \mu_1}{\sigma_1 \sqrt{\nu}} \right) \left(\frac{y - \mu_2}{\sigma_2 \sqrt{\nu}} \right) + \left(\frac{y - \mu_2}{\sigma_2 \sqrt{\nu}} \right)^2}{1 - \rho^2} \right)^{\frac{\nu+2}{2}}}$$

multivariate pearson iv

$$f(\vec{x} \mid \vec{\mu}, \mathbf{\Delta}, \vec{\gamma}, \nu) = \frac{C_N(\mathbf{\Delta}, \vec{\gamma}, \nu) e^{\sum_{i=1}^N \gamma_i \tan^{-1} \left(\frac{x_i - \mu_i}{\sigma_i \sqrt{\nu}} \right)}}{\left(1 + \frac{(\vec{x} - \vec{\mu})^T \mathbf{\Delta}^{-1} (\vec{x} - \vec{\mu})}{\nu} \right)^{\frac{\nu+N}{2}}}$$

Theorem (PIV₁(μ, σ, γ, ν) moments)

The first four central moments (ie. mean, variance, skewness, kurtosis) of PIV₁(μ, σ, γ, ν) can be expressed analytically:

$$\mu_1 = \begin{cases} \mu + \frac{\sigma\gamma\sqrt{\nu}}{\nu-1}, & \nu > 1 \\ \infty, & \nu \leq 1 \end{cases}$$

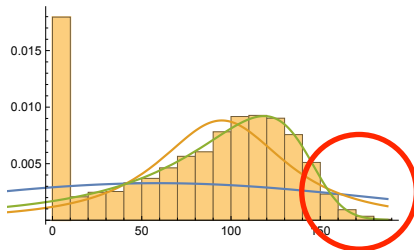
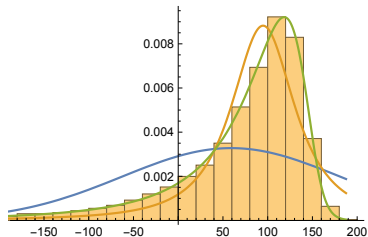
$$\mu_2 = \begin{cases} \left(\frac{\sigma^2\nu}{\nu-2} \right) \left(1 + \left(\frac{\gamma}{\nu-1} \right)^2 \right), & \nu > 2 \\ \infty, & \nu \leq 2 \end{cases}$$

$$\mu_3 = \begin{cases} \frac{4\gamma}{\nu-3} \sqrt{\frac{\nu-2}{\gamma^2 + (\nu-1)^2}}, & \nu > 3 \\ \infty, & \nu \leq 3 \end{cases}$$

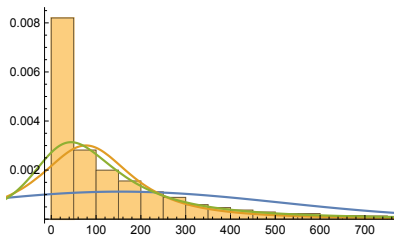
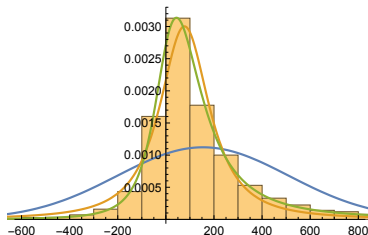
$$\mu_4 = \begin{cases} \left(\frac{3(\nu-2)}{\nu-4} \right) \left(\frac{\gamma^2(\nu+5) + (\nu-3)(\nu-1)^2}{(\nu-3)(\gamma^2 + (\nu-1)^2)} \right), & \nu > 4 \\ \infty, & \nu \leq 4 \end{cases}$$

skewness is created by difference of shape parameters σ_1, σ_2 , causing a *violation of model assumptions*, which explains some invalid p-values seen in the literature [blue: Normal, yellow: t , green: PIV]

LogNormal(5,.1)-LogNormal(4,1)



LogNormal(5,1)-LogNormal(4,1)



- **NEVER SORT BY P-VALUE**
- an experimental result is either significant at a given level or not
- there is no information in a p-value about the *magnitude* of significance
- never claim that something is *more significant* than something else
- report p-values as " $< .001$ " for example, not 10^{-50}

APPENDIX

Noncentrality parameter $\lambda = \sqrt{n} \cdot d$ depends on true shift of test statistic relative to its standard deviation:

$$d = \frac{\mu_1 - \mu}{\sqrt{\mu_2}}$$

From the PIV moments theorem (Murphy, 2010),

$$\Delta\mu = \mu_1 - \mu = \frac{\sigma\gamma\sqrt{\nu}}{\nu-1}, \quad \sqrt{\mu_2} = \frac{\sigma\sqrt{\nu}}{\nu-1} \sqrt{\frac{\gamma^2 + (\nu-1)^2}{\nu-2}}$$

$$\Rightarrow \lambda = \sqrt{n}d = \frac{\sqrt{n} \frac{\cancel{\sigma}\gamma\sqrt{\cancel{\nu}}}{\cancel{\nu-1}}}{\frac{\cancel{\sigma}\sqrt{\cancel{\nu}}}{\cancel{\nu-1}} \sqrt{\frac{\gamma^2 + (\nu-1)^2}{\nu-2}}} = \sqrt{n}\gamma \sqrt{\frac{\nu-2}{\gamma^2 + (\nu-1)^2}} = \sqrt{n} \left(\frac{\nu-3}{4} \right) \mu_3$$

Since the standardized cumulants $\kappa_3 = \mu_3$ and $\kappa_4 = \mu_4 - 3$ are known by the PIV moments theorem and the derivatives of the Normal PDF are

$$\phi^{(k)}(x) = (-1)^k H_k(x) \phi(x),$$

the total bias can be estimated as an Edgeworth expansion of the difference between the true shifted CDF (F) and the nominal standard Normal CDF (Φ):

$$\begin{aligned} \Delta p &= p_{\text{true}} - p_{\text{nominal}} \\ &= (\chi - F(x_c)) - (\chi - \Phi(t_c)) \\ &\approx [\Phi(t_c) - \Phi(x_c)] + \frac{\kappa_3}{6\sqrt{n}} \phi''(x_c) - \frac{1}{n} \left[\frac{\kappa_4}{24} \phi^{(3)}(x_c) + \frac{\kappa_3^2}{72} \phi^{(5)}(x_c) \right] \\ &= [\Phi(t_c) - \Phi(x_c)] + \frac{\mu_3}{6\sqrt{n}} \phi(x_c) H_2(x_c) + \frac{\phi(x_c)}{n} \left[\frac{\mu_4 - 3}{24} H_3(x_c) + \frac{\mu_3^2}{72} H_5(x_c) \right] \end{aligned}$$

Skewness Bias (Δp)

$$\Delta p \approx$$

$$\underbrace{[\Phi(t_c) - \Phi(x_c)]}_{\mathcal{O}(1) \text{ Location Shift}} + \underbrace{\frac{\mu_3}{6\sqrt{n}} \phi(x_c) H_2(x_c)}_{\mathcal{O}(n^{-1/2}) \text{ Skew Correction}} + \underbrace{\frac{\phi(x_c)}{n} \left[\frac{\mu_4 - 3}{24} H_3(x_c) + \frac{\mu_3^2}{72} H_5(x_c) \right]}_{\mathcal{O}(n^{-1}) \text{ Kurtosis \& Variance Corrections}},$$

where $\phi(x)$ and $\Phi(x)$ are the Normal PDF and CDF

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy,$$

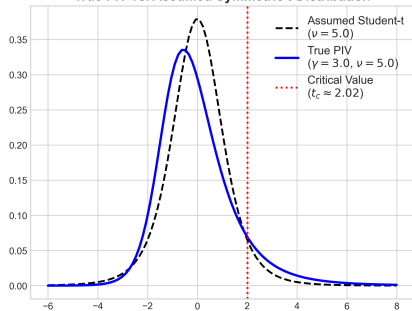
x_c and t_c are the unbiased and biased critical thresholds

$$x_c = t_c - \lambda = t_c - \sqrt{n} \left(\frac{\nu - 3}{4} \right) \mu_3,$$

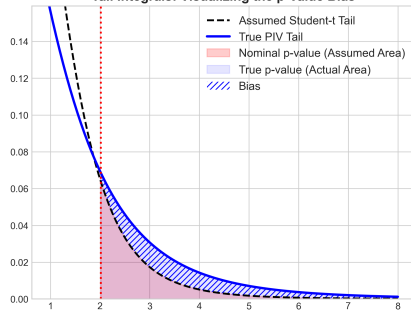
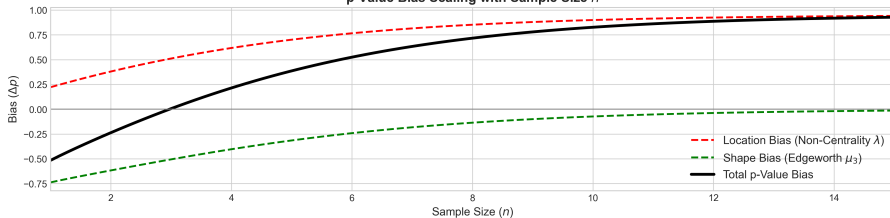
and $H_m(x)$ are the Hermite polynomials

$$H_2(x) = x^2 - 1, \quad H_3(x) = x^3 - 3x, \quad H_5(x) = x^5 - 10x^3 + 15x$$

True PIV vs. Assumed Symmetric t-Distribution



Tail Integrals: Visualizing the p-Value Bias

p-Value Bias Scaling with Sample Size n 

1. The Inherent Asymmetry of Biological Reality

Standard hypothesis testing assumes a perfectly symmetric null distribution ($\mu_3 = 0$). However, physical and biological processes such as conformational energy landscapes, binding affinities, or read counts are often skewed ($\mu_3 \neq 0$).

2. The “Big Data” Amplifier (Computational Discovery)

In high-throughput virtual screening or bioinformatics data mining, sample sizes are massive ($n \sim 10^6$). Because non-centrality term

$$\lambda = \sqrt{n} \left(\frac{\nu - 3}{4} \right) \mu_3$$

scales with $\sqrt{n}$, an infinitesimally small, biologically meaningless skewness is mathematically amplified. Location bias creates pathologically low p -values (eg. $p < 10^{-100}$), prompting a false claim of “extreme statistical significance.”

3. The In-Vitro Replication Failure

An experimental laboratory attempts to validate the computational "discovery" using an *in-vitro* or *in-vivo* assay. However, experimental sample sizes are necessarily small ($n \approx 10$).

Without the $\sqrt{n}$ amplifier, the non-centrality parameter collapses ($\lambda \approx 0$). The identical underlying effect size now yields a non-significant result ($p > 0.05$). The computational finding is declared irreproducible.

Synthesis: Statistical vs. Biological Effect

The exact analytical bias Δp demonstrates a fundamental flaw in applying standard inferential statistics to massive biological skewed datasets. *In the limit as $n \rightarrow \infty$, an extremely small p -value ceases to measure the magnitude of the biological effect, and instead measures the size of the dataset and the skewness of the noise.*